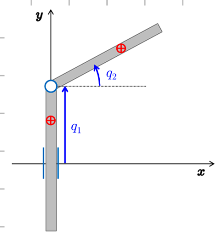
Exercise #1

The PR robot in Fig. 1 moves on a *horizontal* plane. Its inertia matrix has the form

$$M(q) = \begin{pmatrix} A & B\cos q_2 \\ B\cos q_2 & C \end{pmatrix} > 0. \quad (1)$$

Using only the symbolic coefficients A, B and C in (1), provide the expression of the regressor matrix $Y(q, \dot q, \dot q_r, \ddot q_r)$ and of the complete *adaptive* control law that guarantees global asymptotic tracking of a smooth joint trajectory $q_d(t)$, without a priori information on the values of the dynamic coefficients. Neglect all dissipative effects.



The gravity does not produce work on the joints q. We have to compute the Coriolis and centrifugal term:

$$C_1 = \frac{1}{2}\left(\frac{\partial M_1}{\partial q} + \left(\frac{\partial M_1}{\partial q}\right)^T - \frac{\partial M}{\partial q_1}\right) = \begin{pmatrix} 0 & 0 \\ 0 & -B\sin q_2 \end{pmatrix} \Rightarrow \hat c_1 = \dot q^T C_1 \dot q = -\dot q_2^2 B \sin(q_2)$$

model:

$$M\ddot q + \hat c = \tau$$

$$C_2 = \frac{1}{2}\left(\frac{\partial M_2}{\partial q} + \left(\frac{\partial M_2}{\partial q}\right)^T - \frac{\partial M}{\partial q_2}\right) = O \Rightarrow \hat c_2 = 0$$

We let $\dot q_r = \dot q_d + \Lambda(q_d - q)$ where $\Lambda > 0$ (pos. def. 2×2 matrix). Usually $\Lambda = K_D^{-1} K_P$

The control law using $\dot q_r$ to follow q_d is

$$\begin{cases} A\ddot q_{1r} + Bc_2\ddot q_{2r} - \dot q_{2r}\dot q_2 B\sin(q_2) + K_{P1}(q_{d1} - q_1) + K_{D1}(\dot q_{d1} - \dot q_1) = \tau_1 \\ Bc_2\ddot q_{1r} + C\ddot q_{2r} + K_{P2}(q_{d2} - q_2) + K_{D2}(\dot q_{d2} - \dot q_2) = \tau_2 \end{cases}$$

Now i express the linearity in A, B, C

$$\tau = \underbrace{\begin{pmatrix} \ddot q_{1r} & c_2\ddot q_{2r} - \dot q_{2r}\dot q_2 \sin(q_2) & 0 \\ 0 & c_2\ddot q_{1r} & \ddot q_{2r} \end{pmatrix}}_{Y:\ \text{regressor matrix}} \begin{pmatrix} A \\ B \\ C \end{pmatrix} + K_P e + K_D \dot e$$

$\hat e$:= coefficients estimation

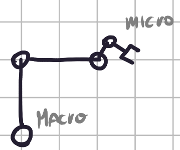
Now we describe the update law for $\hat e$:

$$\hat e = \Gamma \underbrace{\begin{pmatrix} \ddot q_{1r} & 0 \\ c_2\ddot q_{2r} - \dot q_{2r}\dot q_2 s_2 & c_2\ddot q_{1r} \\ 0 & \ddot q_{2r} \end{pmatrix}}_{Y^T} \begin{pmatrix} \dot q_{r1} - \dot q_1 \\ \dot q_{r2} - \dot q_2 \end{pmatrix}$$

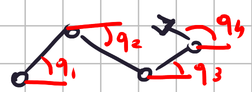
$\Gamma > 0$ is a 3×3 diagonal positive definite matrix.

Exercise #2

A macro-micro planar 4R robot is commanded by kinematic control at the joint velocity level. The first two links have equal length, $L_1 = L_2 = L$, and the last two links are also equal in length but four times shorter, $L_3 = L_4 = L/4$ (micro-manipulator). A trajectory $p_{d,1}(t) \in \mathbb{R}^2$ is assigned to the robot end-effector, which is the highest priority task. A second trajectory $p_{d,2}(t) \in \mathbb{R}^2$ is assigned to the tip of the second link, which is the end of the supporting macro part of the robot. Determine the arm configurations q_s at which *algorithmic* singularities occur for the extended Jacobian of the two simultaneous motion tasks. Specify the additional conditions needed in such algorithmic singularities under which a task priority approach would enable perfect execution of the primary task (with some least-squares error on the secondary task).



we dont use the DH convention. q_i is always the angle between link i and the x asis



the two tasks are:

$$r_1 = \begin{cases} L\left(c_1 + c_2 + \frac{1}{4}[c_3 + c_4]\right) \\ L\left(s_1 + s_2 + \frac{1}{4}[s_3 + s_4]\right) \end{cases} \qquad r_2 = \begin{cases} L(c_1 + c_2) \\ L(s_1 + s_2) \end{cases}$$

the two Jacobian matrices are:

$$J_1 = \begin{pmatrix} -Ls_1 & -Ls_2 & -\frac{L}{4}s_3 & -\frac{L}{4}s_4 \\ Lc_1 & Lc_2 & \frac{L}{4}c_3 & \frac{L}{4}c_4 \end{pmatrix} \qquad J_2 = \begin{pmatrix} -Ls_1 & -Ls_2 & 0 & 0 \\ Lc_1 & Lc_2 & 0 & 0 \end{pmatrix}$$

The augmented Jacobian is:

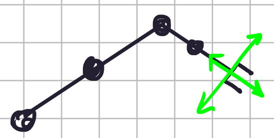
$$J_A = L \begin{bmatrix} -s_1 & -s_2 & -\frac{1}{4}s_3 & -\frac{1}{4}s_4 \\ c_1 & c_2 & \frac{1}{4}c_3 & \frac{1}{4}c_4 \\ -s_1 & -s_2 & 0 & 0 \\ c_1 & c_2 & 0 & 0 \end{bmatrix}$$

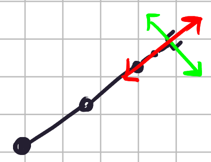
First of all we notice that is singular any time $q_1 = q_2$ (the first two rows becomes identical). Same principle with $q_3 = q_4$. In general

q : singular if $q_1 = q_2 \pm \alpha\pi$, $\alpha \in \{0, 1, -1\}$

$q_3 = q_4 \pm \beta\pi$, $\beta \in \{0, 1, -1\}$ up to $[0, 2\pi]$

The primary task can still be perfectly executed if $q_2 \neq q_3 \pm \gamma\pi$, $\gamma \in \{0, 1, -1\}$





In that case

$$J_A = \begin{array}{c} \left[\begin{array}{cccc} \alpha & \pm\alpha & \gamma & \pm\delta \\ \beta & \pm\beta & \gamma & \pm\delta \\ \alpha & \pm\alpha & 0 & 0 \\ \beta & \pm\beta & 0 & 0 \end{array}\right] \end{array} \begin{array}{l} \left.\begin{array}{l} \\ \\ \end{array}\right\} \text{First two rows non singular} \\ \left.\begin{array}{l} \\ \\ \end{array}\right\} \text{singular} \end{array}$$

Exercise #3

The dynamic model of a robot with n elastic joints interacting with the environment can be expressed by two second-order differential equations, each of dimension n,

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{S}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{\tau}_J + \boldsymbol{J}^T(\boldsymbol{q})\boldsymbol{F} \tag{2}$$

$$\boldsymbol{B}\ddot{\boldsymbol{\theta}} + \boldsymbol{\tau}_J = \boldsymbol{\tau}, \tag{3}$$

named respectively, the *link dynamics* and the *motor dynamics* of the elastic joint robot. In these equations, $\boldsymbol{\theta} \in \mathbb{R}^n$ are the motor variables (before joint elasticity), $\boldsymbol{q} \in \mathbb{R}^n$ are the link variables (after joint elasticity), and $\boldsymbol{\tau}_J = \boldsymbol{K}_J(\boldsymbol{\theta} - \boldsymbol{q}) \in \mathbb{R}^n$ is the elastic joint torque measured by the joint torque sensors, with a diagonal joint stiffness matrix $\boldsymbol{K}_J > 0$. Moreover, the robot is equipped with two encoders per joint, measuring both $\boldsymbol{\theta}$ and $\boldsymbol{q}$. The dynamic terms on the left-hand side of the top equation (2) are the same as in a rigid robot; on the right-hand side, there is also the end-effector robot Jacobian $\boldsymbol{J}(\boldsymbol{q}) = (\partial \boldsymbol{f}(\boldsymbol{q})/\partial \boldsymbol{q})$ associated to the Cartesian task vector $\boldsymbol{x} = \boldsymbol{f}(\boldsymbol{q}) \in \mathbb{R}^m$ and the related generalized Cartesian forces $\boldsymbol{F} \in \mathbb{R}^m$. On the other hand, in the bottom equation (3), the diagonal matrix $\boldsymbol{B} > 0$ collects the reflected motor inertias, while $\boldsymbol{\tau} \in \mathbb{R}^n$ are the input torques available for control. Suppose also that $m = n$.

- Using feedback from joint torque sensors and motor velocities, design first a control law for $\boldsymbol{\tau}$ such that the motor dynamics becomes

$$\boldsymbol{B}_0\ddot{\boldsymbol{\theta}} + \boldsymbol{D}_0\dot{\boldsymbol{\theta}} + \boldsymbol{\tau}_J = \boldsymbol{u}, \qquad \text{for a diagonal, } \textit{arbitrary small } \boldsymbol{B}_0 > 0 \text{ and a suitable } \boldsymbol{D}_0 > 0.$$

First of all, we have to apply feedback linearization on the motor side:

$\tau = \tau_J + B a \Rightarrow \ddot{\theta} = a$ now i choose

$\ddot{\theta} = \underbrace{B_0^{-1}(u - D_0\dot{\theta} - \tau_J)}_{a} \Rightarrow$

$B_0\ddot{\theta} = u - D_0\dot{\theta} - \tau_J \Rightarrow B_0\ddot{\theta} + D\dot{\theta} + \tau_J = u$ so the law is

$\tau_J + B B_0^{-1}(u - D_0\dot{\theta} - \tau_J) = B B_0^{-1} u - B B_0^{-1} D_0 \dot{\theta} + (I - B B_0^{-1})\tau_J$

- Thanks to this inertial reduction, the motor dynamics is made arbitrarily fast so that we can assume $\boldsymbol{\tau}_J \simeq \boldsymbol{u}$ (this fast dynamics can be seen as if it were always at steady-state). Complete then the control design on the robot link dynamics by imposing to the Cartesian task vector $\boldsymbol{x}$, the following impedance model

$$\boldsymbol{M}_x(\boldsymbol{q})\ddot{\boldsymbol{x}} + (\boldsymbol{S}_x(\boldsymbol{q},\dot{\boldsymbol{q}}) + \boldsymbol{D}_m)\dot{\boldsymbol{x}} + \boldsymbol{K}_m(\boldsymbol{x} - \boldsymbol{x}_d) = \boldsymbol{F} \tag{4}$$

where $\boldsymbol{x}_d$ is constant, $\boldsymbol{K}_m > 0$ and $\boldsymbol{D}_m > 0$ are desired, diagonal stiffness and damping matrices, and, assuming to work out of kinematic singularities,

$$\boldsymbol{M}_x(\boldsymbol{q}) = \left(\boldsymbol{J}(\boldsymbol{q})\boldsymbol{M}^{-1}(\boldsymbol{q})\boldsymbol{J}^T(\boldsymbol{q})\right)^{-1}, \quad \boldsymbol{S}_x(\boldsymbol{q},\dot{\boldsymbol{q}}) = \boldsymbol{J}^{-T}(\boldsymbol{q})\boldsymbol{S}(\boldsymbol{q},\dot{\boldsymbol{q}})\boldsymbol{J}^{-1}(\boldsymbol{q}) - \boldsymbol{M}_x(\boldsymbol{q})\dot{\boldsymbol{J}}(\boldsymbol{q})\boldsymbol{J}^{-1}(\boldsymbol{q}).$$

We have that $\tau_J \simeq u$, so we assume that we can control directly τ_J. the link dynamics then becomes:

$M\ddot{q} + S\dot{q} + g = u + J^T F$ At this point we have to describe this model in the cartesian space:

$M_x\ddot{x} + S_x\dot{x} + g_x = J^{-T} u + F$, now we have to choose u to impose the impedance model.

$$M_x \ddot{x} + S_x \dot{x} + g_x = J^T u + F$$

We apply FBL: $u = J^T (M_x a + S_x \dot{x} + g_x - F) \Rightarrow$

$\ddot{x} = a$ So we let

$$a = M_x^{-1}\left(-(S_x + D_m)\dot{x} + K_m(x_d - x) + F\right) \Rightarrow$$ guarantees the impedance model.

The joint control input u becomes

$$u = M J^{-1}\left(-\dot{J}\dot{q} + M_x^{-1}\left(D_m(-\dot{x}) + K_m(x_d - x)\right)\right)$$

$+ c + g$ notice how no Force measurement is required.

Since the control torque is $\tau = BB_0^{-1}u - BB_0^{-1}D_0\dot{\theta} + (I - BB_0^{-1})\tau_J$

the COMPLETE control law is: $\dot{x} = J\dot{q}$ $\quad K(q)$ DK

$$\tau = BB_0^{-1}\left[M J^{-1}\left(-\dot{J}\dot{q} + M_x^{-1}\left(D_m(-\dot{x}) + K_m(x_d - x)\right)\right) + S\dot{q} + g\right] - BB_0^{-1}D_0\dot{\theta} + (I - BB_0^{-1})\tau_J$$

notice how is based only on the known model and on the state variables. The corresponding behaviour is

Write the final control law for the input torque $\boldsymbol{\tau} \in \mathbb{R}^n$ in explicit form only in terms of the original state variables $\boldsymbol{q}$, $\dot{\boldsymbol{q}}$, $\boldsymbol{\theta}$, and $\dot{\boldsymbol{\theta}}$. Moreover, if an external constant force $\boldsymbol{F} = \bar{\boldsymbol{F}}$ is being applied from the environment to the robot, which will the equilibrium $\boldsymbol{x} = \boldsymbol{x}_E$ at steady state and what will be the value $\boldsymbol{\tau}_E$ of the control torque $\boldsymbol{\tau}$ at this equilibrium?

$$M_x \ddot{x} + (S_x + D_m)\dot{x} + K_m(x - x_d) = \bar{F}$$ i denote $B_m = S_x + D_m$

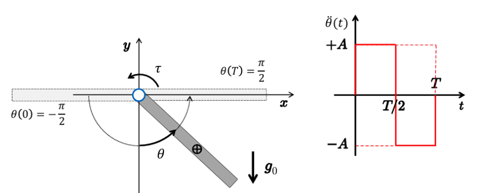
$$M_x \ddot{x} + B_m \dot{x} + K_m(x - x_d) = \bar{F}$$ at steady state we have

$$K_m(x - x_d) = \bar{F}$$ So the Final conf. will be

$$x_E = K_m^{-1}\bar{F} + x_d$$

$$\tau_E = BB_0^{-1} M J^{-1}\left(-M_x^{-1}\bar{F} + g\right) + (I - BB_0^{-1})\tau_J$$

Exercise #4



Consider the actuated link under gravity in Fig. 2, with the input torque bounded as $|\tau| \le \tau_{max}$. The link should perform a rest-to-rest motion from $\theta(0) = -\pi/2$ to $\theta(T) = \pi/2$ with the bang-bang acceleration profile $\ddot{\theta}(t)$ shown in the same figure. Determine analytically the minimum feasible time T to execute the desired motion with this type of trajectory.

let L to be the distance between the joint and the CoM.

$$p = \begin{cases} L\sin\theta \\ -L\cos\theta \end{cases} \Rightarrow \dot{p} = \begin{cases} L\cos\theta\,\dot{\theta} \\ L\sin\theta\,\dot{\theta} \end{cases} \qquad \dot{p}^T\dot{p} = L^2\dot{\theta}^2 \Rightarrow T = \frac{1}{2}mL^2\dot{\theta}^2 + \frac{1}{2}I\dot{\theta}^2 = \frac{1}{2}\dot{\theta}^2(I + mL^2)$$

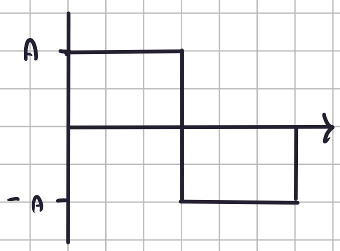
$$U = -mg_0L\cos\theta \Rightarrow \frac{\partial U}{\partial \theta} = mg_0L\sin\theta$$ the dyn. model is:

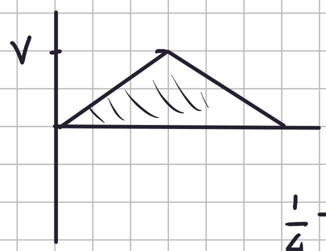
$$\underbrace{(I + mL^2)}_{e_1}\ddot{\theta} + \underbrace{mg_0L}_{e_2}\sin\theta = \tau \Rightarrow e_1\ddot{\theta} + e_2\sin\theta = \tau$$

In $t = T$ the motor will have to exert $-A \frac{rad}{sec^2}$. In that conf, $\theta = \frac{\pi}{2}$ so: $e_1 \ddot{\theta} + e_2 = \tau \Rightarrow |\ddot{\theta}| = \left|\frac{1}{e_1}(\tau - e_2)\right| = A \Rightarrow$

$\sin\theta = 1$

$\frac{1}{e_1}(\tau_{max} - e_2) = A$ The profile is





$V = \frac{T}{2}A$ total displ. is π

$\pi = \frac{T}{2}V = \frac{T}{2}\frac{T}{2}A = \frac{T^2}{4}A = \pi$

$\frac{1}{4}T^2 \frac{1}{e_1}(\tau_{max} - e_2) = \pi \Rightarrow T = \sqrt{\frac{4\pi e_1}{\tau_{max} - e_2}}$